

Ethical and Legal Assessment

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Ethical and Legal Assessment

1. Ethical Considerations

1.1 Data Privacy

The project will strictly adhere to data privacy principles consistent with the UK General Data Protection Regulation (GDPR) and NHFT Information Governance (IG) standards.

- **Data Privacy Principles:** The use of data will comply with purpose limitation and data minimisation. Data are used solely for operational modelling and service planning. Transparency will be maintained by clearly documenting data sources, processing steps, and intended uses.
- **Data Ownership:** NHFT remains the data controller and retains ownership of all datasets accessed and processed. The data will be used under existing organisational permissions within the Business Intelligence (BI) function.
- **Sensitive Data:** The project involves potentially sensitive healthcare activity data (referrals, appointments, and workforce information). All identifiable data will be pseudonymised before analysis. Access will be restricted to authorised staff working within NHFT's secure data environment, with audit logs to track all data interactions.

1.2 Fairness and Bias

Ensuring fairness and mitigating bias are core priorities of the modelling design.

- **Bias Identification:** Potential bias could arise from inconsistent data quality across services or underrepresentation of smaller or specialist services. Where potential bias is detected, tweaks to the model and parameter configurations will be applied, such as adjusting data input selections, weighting schemes, or algorithm parameters to improve fairness and representativeness across service types.
- **Fairness Metrics:** The fairness of forecasts will be assessed using comparative error metrics (e.g., variance in error rates across services or divisions) to ensure balanced accuracy and prevent overfitting to dominant service groups.
- **Algorithmic Transparency:** All algorithms used (e.g., ARIMA, ETS, Random Forest) will be documented, with model assumptions and limitations clearly explained. Outputs will include interpretability features (e.g., feature importance, residual analysis) to ensure models are understandable and auditable by non-technical stakeholders.

1.3 Accountability and Responsibility

Clear accountability and governance frameworks will be in place throughout the project lifecycle.

- **Accountability Frameworks:** Roles and responsibilities will be defined in the project plan, including those of the Data Scientist (project lead), BI Manager (data access oversight), and Performance Improvement Manager (project manager).
- **Ethics Committees:** As the project involves patient-related operational data, ethical oversight will be provided through NHFT's Information Governance and Data Protection teams. Any amendments affecting data use or scope will be reviewed through the Trust's internal IG processes.

2. Navigating Legal Requirements

2.1 Data Protection Regulations

The project complies with data protection laws governing the handling of healthcare data.

- **GDPR:** The project aligns with UK GDPR principles, ensuring lawfulness, fairness, and transparency. No new data will be collected; existing operational data will be repurposed with proper justification. A Data Protection Impact Assessment (DPIA) will be completed before data processing begins.
- **HIPAA:** Although HIPAA is a US regulation, equivalent safeguards are observed in line with NHS confidentiality standards and the Caldicott Principles, ensuring patient-level data is not exposed or misused.

2.2 Intellectual Property

Ownership and licensing are clarified from the outset.

- **Copyright and Licensing:** All datasets remain the property of NHFT. Analytical scripts, models, and dashboards developed as part of this project will be owned by NHFT but may be shared with the University of Nottingham for assessment purposes under the apprenticeship agreement.
- **Open-Source Compliance:** Open-source Python libraries (e.g., Pandas, Scikit-learn, Statsmodels, Dash) will be used under their respective licences. All packages will be referenced appropriately, and no restricted or commercial libraries will be used without permission.

2.3 Data Retention and Destruction

Appropriate data lifecycle management practices will be applied.

- **Data Retention Policies:** Data will be retained only for the duration of the project and any agreed retention period under NHFT's Records Management Code of Practice.
- **Secure Data Destruction:** Upon project completion, temporary or local copies of pseudonymised data will be deleted securely using NHFT-approved methods (e.g., encrypted deletion or secure network purging). The project will also aim to develop an automated, direct-query data process where models access data dynamically from trusted BI sources without creating intermediate storage files. This approach minimises the need to retain data externally and further reduces risk of data exposure.

3. Best Practices

3.1 Data Governance

Strong governance underpins the project's credibility and compliance.

- **Data Catalogues:** A structured data inventory will document data sources, update frequency, data owners, and access rights. Metadata will include field descriptions, update dates, and version tracking.
- **Data Security:** Data will remain on NHFT's secure network with access controlled through role-based permissions. Encryption in transit (SSL) and at rest will be ensured by NHFT IT infrastructure. Regular penetration testing and audits are conducted by NHFT ICT to maintain security assurance.

3.2 Informed Consent

- **Informed Consent Forms:** As this project uses existing operational data collected for service delivery, no direct patient consent is required.
- **Consent Management:** Any future use involving identifiable or new data collection will require explicit patient consent and review through NHFT's IG processes.

3.3 Documentation and Transparency

Transparency ensures that outcomes are explainable and reproducible.

- **Project Documentation:** Comprehensive documentation will be maintained through Git (codebase), OneNote (logbook), and SharePoint (technical documentation). This includes data lineage, preprocessing steps, model design decisions, and validation results.
- **Transparency Reports:** Final reports and dashboards will explain model assumptions, potential biases, and limitations in clear, accessible language. This supports informed decision-making by stakeholders and aligns with NHS England's Data Ethics Framework principles.

Summary

This framework ensures that all data handling and modelling activities respect privacy, fairness, accountability, and the wider societal value of data science in improving patient care and service delivery. Ethical and legal compliance will be monitored continuously through audit trails, regular reviews, and supervisor feedback.

Ethical Considerations

Theme	Considerations / Details in Relation to the Project
Seek to enhance the value of data science for society	The project aims to improve healthcare service planning and delivery across NHFT's services by providing data-driven insights into demand, capacity, and resource utilisation. By enabling more efficient workforce allocation and reducing patient waiting times, the platform directly supports better access to care and overall patient well-being. The reusable and scalable nature of the model enhances organisational efficiency and contributes positively to the wider Northamptonshire NHS system.
Avoid harm	All data used in the project will be pseudonymised, stored, and processed within NHFT's secure data environment. The model will comply with GDPR, Data Protection Act 2018, and NHS IG standards. No individual-level patient decisions will be automated without human oversight. Ethical risks such as potential bias in forecasting models (e.g. services with smaller populations or inconsistent reporting) will be identified and mitigated through fairness checks and transparent documentation.
Apply and maintain professional competence	The project applies established best practices in statistical modelling, data engineering, and ethical data handling. All methods and tools (e.g., Python, SQL, Dash) will follow reproducible coding standards, version control, and validation protocols. Ongoing review with NHFT BI colleagues and academic supervisors will ensure compliance with professional and legal standards while maintaining analytical integrity.
Seek to preserve or increase trustworthiness	Transparency will be embedded in the project design through open documentation of model assumptions, limitations, and data sources. Stakeholders, including operational leads and service managers, will be engaged throughout the design and testing phases to ensure outputs are interpretable and aligned with operational needs. Findings will be communicated clearly through dashboards and presentations, using non-technical explanations where necessary to build understanding and trust.

Maintain accountability and oversight	The project will be governed under NHFT's Information Governance and Research & Development frameworks, with approvals sought as required. Oversight will include internal BI leadership and academic supervision. All major design decisions will be logged in the project's version control and OneNote record. Ethical considerations, data protection, and audit trails will be reviewed at regular project checkpoints to ensure continued compliance and accountability.
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Legal Issues

Theme	Considerations / Details in Relation to the Project
Data Protection and Privacy Laws	The project will comply with the UK GDPR, Data Protection Act 2018, and NHS Digital's Data Security and Protection Toolkit standards. All identifiable data will remain within NHFT's secure network. Access will be limited to authorised staff with appropriate IG training. The project does not involve direct patient contact or consent gathering, as data are collected for operational purposes. Appropriate technical measures such as encryption, password protection, and secure server storage will be implemented to prevent unauthorised access or data breaches.
Data Anonymisation and Pseudonymisation	All datasets will be pseudonymised prior to analysis by removing or replacing patient identifiers with unique, non-identifiable codes. No direct identifiers (e.g., names, NHS numbers, addresses) will be used. Aggregated results will be produced for reporting, ensuring individuals cannot be re-identified through small-number disclosure. Anonymisation will follow NHS guidance and best practice to minimise re-identification risk.
Data Ownership and Intellectual Property	NHFT is the data controller and retains ownership of all data used in the project. The models, code, and outputs developed will be considered NHFT intellectual property, although academic elements may be shared with the University of Nottingham under the apprenticeship agreement. Any open-source components will be used in accordance with their licensing terms, ensuring no breach of commercial restrictions.
Data Sharing and Third-Party Agreements	Data used for this project will not be shared externally. If collaboration with the University of Nottingham requires demonstration or review of outputs, only non-disclosive, anonymised summaries will be provided. Any future sharing with partner organisations (e.g., ICS or NHS England) would require a formal Data Sharing Agreement (DSA) and approval from NHFT's Information Governance and Caldicott Guardian teams.
Compliance with Sector Specific Regulations	The project will adhere to NHS England and NHS Digital data governance frameworks, including the Caldicott Principles. As the data involve healthcare activity, compliance with health-specific regulations (e.g., NHS Records Management Code of Practice, confidentiality standards) is mandatory. The project will undergo a Data Protection Impact Assessment (DPIA) to ensure appropriate safeguards are in place before accessing or processing data.

Documentation and Audit Trail	A comprehensive audit trail will be maintained through version control (Git) and the project logbook (OneNote). All datasets, transformations, models, and access requests will be documented and time-stamped. The project will maintain evidence of data lineage, ensuring reproducibility and transparency for internal audit or external review. This also supports accountability and traceability in accordance with NHS IG standards.
Algorithmic Accountability and Transparency	All modelling processes (forecasting, optimisation, and simulation) will be fully documented, including parameters, assumptions, and evaluation metrics. Outputs will be interpretable and accompanied by explanations of uncertainty and model limitations. No automated decision-making affecting individuals will occur without human oversight. The project will ensure fairness by testing models across different service types to avoid systematic bias. Transparency in design and communication will preserve trust and compliance with ethical AI principles.